FraudGraph AI — Project Overview & Academic Report

Adaptive Financial Fraud Detection Under Concept Drift with Graph Analytics & Epistemic Uncertainty

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PROJECT GUIDE
Guide Teacher / Supervisor

TECHNICAL DOMAIN
FinTech & Applied ML

EVALUATION STATUS
Verified & Complete

1. EXECUTIVE SUMMARY & PROBLEM MOTIVATION

Modern digital banking rails process millions of transactions per second, rendering traditional Anti-Money Laundering (AML) monitoring systems increasingly ineffective. Traditional anti-fraud architectures fail due to three critical structural limitations:

- **Coordinated Mule Rings (Structuring & Layering):** Fraud syndicates distribute illicit transfers across decentralized cyclic networks ($A \to B \to C \to A$) to bypass isolated account-level thresholds.
- **Adversarial Concept Drift:** Criminal syndicates rapidly alter payment channels, transaction velocities, and spend amounts to evade stationary machine learning classifiers.
- **Investigation Fatigue:** False positive rates in legacy rules often exceed 90%, overwhelming compliance teams and delaying high-priority fraud freezes.

FraudGraph AI resolves these bottlenecks through a hybrid architecture combining **directed graph topology analytics**, **dynamic spend deviation profiling**, and **epistemic model uncertainty estimation**. It autonomously blocks high-confidence attacks, approves clean transactions, and prioritizes drifted/ambiguous edge cases for active-learning human review.

2. HIGH-LEVEL SYSTEM ARCHITECTURE

1. Client & Presentation Layer

Built with **React 18**, **Vite 5**, **TailwindCSS**, and **Recharts**. Features an interactive SVG topology explorer, sub-second live ingestion feed, KPI dashboards, and investigator drawers.

2. Ingestion & Security Gateway

Powered by **Java 17/25** & **Spring Boot 3.3.4**. Implements **Spring Security 6** with stateless **JJWT** bearer authentication, BCrypt password encryption, and granular RBAC.

3. Real-Time Detection Engines

In-memory sliding-window multi-graphs, depth-bounded DFS cycle detection, dynamic spend Z-scores, and statistical Page-Hinkley distribution distance estimators.

4. Persistence & Machine Learning

Dual-profile persistence (H2 in-memory for instant development, MySQL 8 for production). Backed by Python research benchmarks on 5,000 synthetic transactions.

3. CORE ALGORITHMIC FORMULATIONS

A. Depth-Bounded Graph DFS Cycle Detection

To detect money laundering circular layering loops ($A \to B \to C \to A$) in real time without exponential path explosion, an in-memory directed multigraph maintains recent edges and applies Depth-First Search with depth constraint $k \le 4$:

GRAPH TOPOLOGY DETECTION FORMULAS

$$\text{Cycle}(u, v) = \text{DFS}(v, \text{target} = u, \text{depth} \le 4) \implies \text{Circular Layering (+45 pts, CRITICAL)}$$

Accumulator Mule (Fan-In): $\deg_{\text{in}}(v) \ge 3$ | Disperser Mule (Fan-Out): $\deg_{\text{out}}(u) \ge 3$ (+35 pts, HIGH)

B. Spend Deviation Z-Score

BEHAVIORAL SPEND OUTLIER METRIC

$$Z = (\text{Amount} - \mu_{\text{baseline}}) / \sigma_{\text{baseline}} \implies \text{If } Z > 2.5, \text{ Flag Spend Outlier (+25 pts)}$$

3. CORE ALGORITHMIC FORMULATIONS (CONTINUED)

C. Model Epistemic Uncertainty Calculus

Traditional models output only a binary probability P. FraudGraph AI calculates a dual-axis metric: risk score and **epistemic uncertainty ($U \in [0.02, 0.98]$)**. Uncertainty peaks at the decision boundary ($P = 0.50$) and escalates when feature dissonance or concept drift occurs:

DUAL-FACTOR EPISTEMIC UNCERTAINTY FORMULATION

$$U = \max(0.02, \min(0.98, (1.0 - 2 \times |P - 0.5|) \times 0.55 + \delta_{\text{conflict}} + 0.25 \times D_{\text{drift}}))$$

where $\delta_{\text{conflict}} = 0.20$ when behavioral spend and graph topology signals contradict each other.

D. Active Learning Prioritization Formula

INVESTIGATOR QUEUE PRIORITY METRIC

$$\text{PriorityScore} = 0.60 \times \text{RiskScore} + 40.0 \times U$$

This formula ensures compliance teams prioritize transactions that are both high risk and near the model's ambiguity boundary, maximizing active-learning retraining efficiency.

4. COMPLETE TECHNOLOGY STACK

LAYER	TECHNOLOGY	VERSION	ROLE & CAPABILITIES IN PROJECT
Backend	Java / Spring Boot	Java 17+, Boot 3.3.4	REST APIs, stateless JWT filter, transactional JPA repos, scheduled simulator.
Security	Spring Security / JJWT	Security 6.3, JJWT 0.12	BCrypt password hashing, stateless token validation, investigator/admin RBAC.
Frontend	React / Vite / Tailwind	React 18.3, Vite 5.4	High-density dark mode SPA, responsive drawers, modal dialogs, status badges.
Charts & Graph	Recharts / Lucide / SVG	Recharts 2.12	Streaming Area charts, Priority Bar series, Concept Drift Line, SVG topology map.
Database	H2 / MySQL	H2 (Dev) / MySQL 8.0	Embedded in-memory zero-config dev DB; production schema with foreign keys and indexes.
ML Suite	Python 3 / scikit-learn	Python 3.10+	Synthetic dataset generator (5,000 txns), sliding-window drift benchmark.

5. FUNCTIONAL MODULES & IMPLEMENTATION BREAKDOWN

1. Executive Dashboard
KPI metrics for volume, fraud rate %, latency, Recharts risk density area series, and drift threshold lines.

2. Ingestion Stream
Live transaction feed with status pills (APPROVED , REVIEW , BLOCKED) and synthetic custom injection.

3. Topology Explorer
Interactive SVG visualizer displaying circular money laundering rings in glowing red and mule hubs in amber.

4. Case Management & Explainable AI (XAI)
Priority-sorted queue displaying factor waterfall attributions (+35 Loop, +30 Velocity). Supports complete case resolutions (CONFIRMED_FRAUD , FALSE_POSITIVE , DISMISSED) with immutable timestamped audit notes and 1-click CSV export.

5. Attack Simulator & Drift Workbench
Stress lab allowing compliance officers to inject synthetic fraud attacks: Circular Mule Rings, Velocity Smurfing Bursts, and Inflationary Drift Transfers with continuous background streaming toggle.

6. EMPIRICAL RESEARCH & ML BENCHMARK RESULTS

The research module evaluated an end-to-end synthetic dataset of **5,000 transactions** with extreme class imbalance (3.4% fraud rate) and temporal concept drift introduced at record #3,500:

Key Benchmark Findings:

- **Real-Time Latency:** Spring Boot processing latency averaged **14.5 milliseconds** per transaction, fully satisfying real-time payment SLAs.
- **Pre-Drift Window (Records 1 – 3,500):** Precision: 14.1%, Recall: 65.2%, Mean Model Uncertainty: **0.4346**.
- **Post-Drift Window (Records 3,501 – 5,000):** Recall: 72.7%, Mean Model Uncertainty: **0.5384 (+0.1038 shift)**.
- **Empirical Proof:** The statistically significant rise in epistemic uncertainty ($+0.1038$) successfully flagged newly drifted fraud topologies for human review without catastrophic false negative spikes.
- **Drift Early Warning:** The Page-Hinkley streaming detector triggered its first alert at record #957 ($D = 0.4127 \ge 0.35$).

7. COMPLIANCE, ETHICS & REGULATORY SAFEGUARDS

- **Zero Sensitive PII / Banking Records:** The platform relies 100% on synthetic mathematical distributions (log-normal spend, synthetic account numbers). Zero unauthorized consumer data is used.
- **Regulatory XAI Standard:** In compliance with **GDPR Article 22** (Right to Explanation) and the **US Fair Credit Reporting Act (FCRA)**, every blocked transaction generates an auditable, human-readable JSON breakdown of contributing factors.

8. PROJECT VERIFICATION & DELIVERABLES CHECKLIST

DELIVERABLE	VERIFICATION STATUS	EVIDENCE / OUTPUT
Spring Boot 3 Backend	VERIFIED	All 6/6 unit & integration tests passing (`mvn test`); live on port 8080.
React 18 Frontend	VERIFIED	Clean production build (`vite build`); interactive SPA on port 5173.
ML Research Suite	VERIFIED	All 3 benchmark scripts run via `npm run ml:benchmark`.
One-Click Startup Script	VERIFIED	`start.sh` auto-detects Java 17/25, launches background servers cleanly.

9. ACADEMIC APPROVAL & GUIDE SIGN-OFF

This document summarizes the system architecture, mathematical formulations, software implementation, and empirical research benchmarks completed for **FraudGraph AI**.

CANDIDATE / STUDENT SUBMISSION

P. Tattvamasi
Candidate Signature

Date: November 2, 2026 Status: Submitted

PROJECT GUIDE / SUPERVISOR REVIEW

Guide Teacher Name
Supervisor Signature & Remarks

Date: ____ / ____ / 2026 Grade / Approval: _____